

Sea of Japan / East Sea model validation

Field	Entry
Region	Sea of Japan / East Sea, LME_050; Sea Around Us target boundary. Regional workbook
Catch source	Sea Around Us v50.1, 1950–2019. Reference: 2019 landings, including reported and unreported retained catch; discards separate. Raw catch Taxon/TL source
Selected article	Inoue, Watari, Sawada, Lavergne & Yamashita (2023), Impacts of regime shift on the fishery ecosystem in the coastal area of Kyoto prefecture, Sea of Japan, assessed using the Ecopath model. Fisheries Science 89:573–593; DOI 10.1007/s12562-023-01691-9. Main paper Publisher supplement
Other known articles	No other candidate article is recorded for this selection. The 2024 publisher correction concerns Open Choice/copyright; it supplies no numeric or year-specific diet correction. Publisher correction
Selected model	50_502013_Coastal_Kyoto_Inoue_(2013); static 2013 coastal Kyoto model, reported 2,230 km ² from the coastline to 240 m depth. 40 ecological groups plus synthetic diet_import (41). Selected JSON
Other models from the same article	50_501985_Coastal_Kyoto_Inoue_(1985), cold-regime counterpart, remains an explicitly good comparison candidate. Both years use the shared, year-ambiguous published diet matrix. 1985 JSON
Selection rationale	Article: “only available paper.” Model: the user chose 2013 because it is newer; 1985 is also good for comparison. Recency does not demonstrate numerical superiority or whole-LME suitability. Recorded rationale
Model extraction	All 240 final Table 1 source cells match accepted 2013 values. One Table 2 diet matrix is captioned both years despite year-specific sardine/anchovy prose. Nine nonunit diet columns, accepted normalization, GS 0.2, solved accumulation, missing/default catch/migration and single-detritus routing remain documented. Supplementary depth-bin areas total 1,223 km ² versus main 2,230 km ² ; no scientific repair, rescaling or new normalization. Source review Source cells
SPPR calculation	[MANUAL — Researcher: calculation choices and reasons] [Excluded groups, scope and relevant values; other adjustments]

Field	Entry
GE diagnostics	No negative SPPR entries in import (41), Detritus (40) or Phytoplankton (39) columns across all 41 groups, including unfished groups. rho_living=0.10604454; b=0.03129443; Detritus (40) SPPR=1.36109201.
TE diagnostics	No negative SPPR entries in import (41), Detritus (40) or Phytoplankton (39) columns across all 41 groups, including unfished groups. rho_living=0.24614202; Detritus (40) SPPR=0.67578005.
Geographic fit	Approximate A: 0.15–0.23% of the target region. B: 95–100% of the study area inside the target. Indicative ranges; graphical trace differs from the reported area.
Temporal fit	Static 2013 coefficients and source catch/biomass splits are transferred across 1950–2019. The model postdates the 1980s regime shift; fixed Kyoto guild composition does not describe observed whole-LME change across years, fleets or catch bases.
Other	Historical GE, TE and With Egestion diagnoses remain WARN; strict balance passes. Tongue sole (30) EE is 0. Every source-recipient entry is finite and nonnegative. Exact old/current loader parity supports reuse; no fresh solve was performed. Production eligibility remains false. Full matrices Loader parity
Open issues and next action	[Researcher: open issues and next actions]
Review and reproducibility	Researcher name: [name] review date: [date]

Taxon mapping and coverage

Reference: 2019 landings; all exact catch labels, including unidentified and zero-catch records; no taxon or group filter.

40 ecological source groups (+1 computational import); 253 catch labels.

Simple-chain PPR totals 145,016,301.85 t C in 2019.

35 labels lack a saved classic coefficient/TL; all have zero 2019 landings and zero annual PPR. Unavailable TL remains ?, not zero.

Method: independent classic/simple trophic chain using the saved taxon coefficient and a single carbon conversion.

[Full taxon mapping appendix](#) and [descriptive Sources sheet](#)

Overall confidence	Taxa	Catch percentage	PPR percentage
High	20	28.4481%	20.1996%
Medium	89	18.5567%	18.5625%
Low	1	0.0124%	0.0178%
Very low	143	52.9828%	61.2201%
Unresolved	0	0.0000%	0.0000%

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Closest represented taxonomic or ecological analogue for an absent compartment	Very low	56.7420%
Taxon explicitly named in the selected source group	High	19.0223%
Related unlisted taxon extends a representative source pool	Medium	10.3494%
Represented candidate set established with a stated regional extension	Medium	8.2132%
Broad reporting category approximated by its supported source pools	Very low	4.4780%
Authoritative synonym connects the source name to the catch label	High	1.1773%
Partial/conflicting source identity or historical reporting scope	Low	0.0178%

Membership evidence: [Table S3 and source methods](#) | [all taxon decisions](#). Representative lists are not exhaustive. Residual bony fish exclude Stingray; source pelagic fish include Red seabream and Seabass. Name joins resolve the S3 Sole/Dragonet row reversal.

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
One candidate receives the entire taxon catch	High	72.3367%
Complete source-model catch proportions assumed for regional catch composition	Medium	23.1821%
Complete source-model biomass proportions assumed after source catch is incomplete	Medium	4.4812%

Overall confidence is the weaker membership/allocation component. Complete 2013 S4 detailed fleet catches supply Medium split proxies; incomplete catch sets use complete final Table 1 biomass with common catchability assumed. Missing source rows and loaded zeros are distinct. Both proxy types transfer local historical composition across regional years, fleets and bases; taxon-specific observed shares were unavailable.

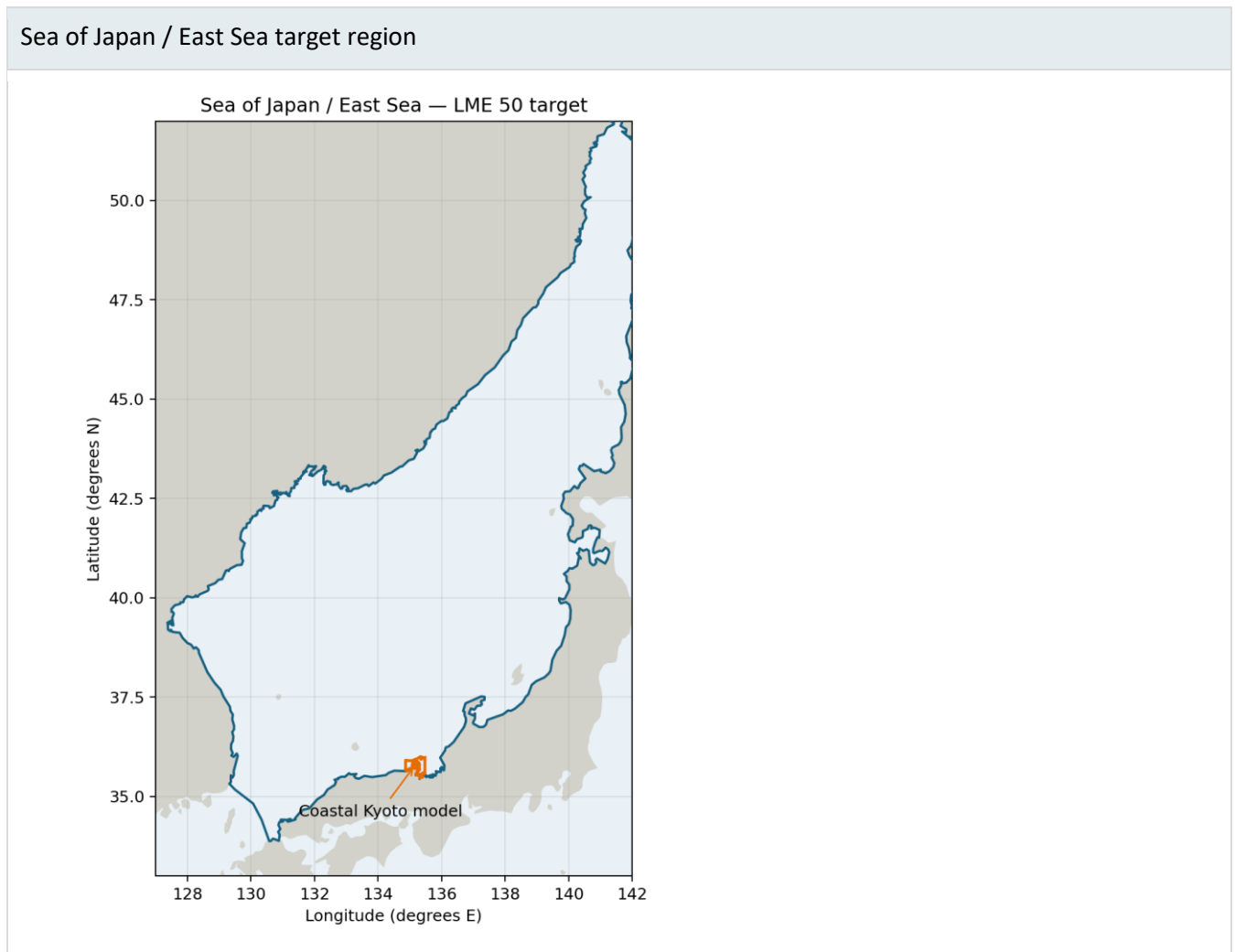
[Allocation evidence](#)

No unresolved labels remain. Very low: 143 labels, 52.9828% of landings and 61.2201% of classic PPR. Absent northern/deep fish, salmon, sharks, broad residuals and jellyfish retain explicit analogue limits. This cohort contributes 66.52% of GE, 76.86% of TE and 64.39% of With Egestion PPR under the same controls; assignment completeness does not establish scientific suitability. [Method-specific contributions](#)

Geographic evidence screenshots

R is the Sea Around Us Sea of Japan / East Sea LME target. S follows the dashed model boundary in source Figure 1, from the coastline to 240 m. Native longitude/latitude ticks calibrate the curved offshore edge and source coastline/islands; 12 water/land controls pass. The historical broad rectangle is rejected. The trace remains approximate and does not establish exact bathymetry.

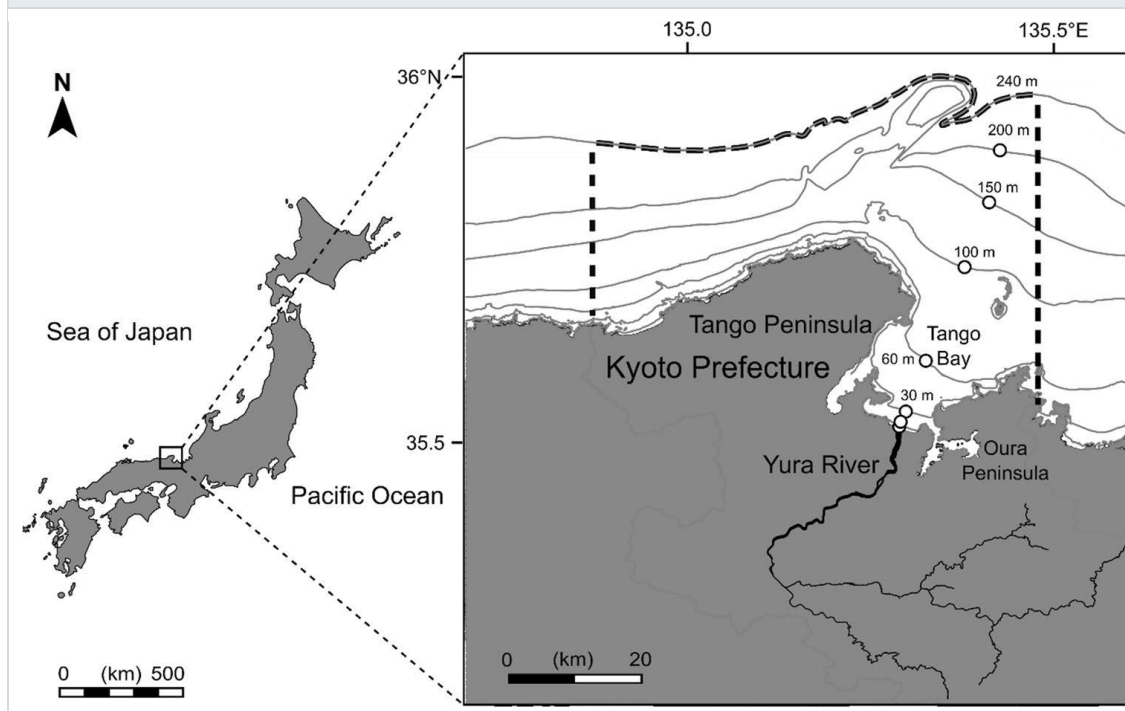
Region area



Sea Around Us LME 50 target R: approximately 988,290 km². Orange marks the small coastal Kyoto model location. [Target boundary](#) | [Map image](#)

Study area in the article

Original coastal Kyoto model area — Figure 1



Inoue et al. 2023 Figure 1, PDF 2 / printed p.574; methods PDF 4 / p.576: model domain is coastline to 240 m, reported 2,230 km². The dashed border is the model area; white circles are sampling stations. Figure reproduced without geographic alteration; article CC BY 4.0. [Original figure](#) | [Methods](#) | [CC BY 4.0](#)

Approximate trace S is 1,681 km² and intersection 1,654 km²: central graphical A≈0.17%, B≈98%. The trace is 24.62% below the reported 2,230 km², an unresolved discrepancy larger than graphical sensitivity. Indicative Low ranges A≈0.15–0.23%, B≈95–100% retain both uncertainties; they are not statistical intervals. Supplementary S1 depth-bin areas total 1,223 km² without reconciliation to 2,230 km²; this is a separate benthic biomass-denominator limit, not a replacement polygon. Orange is the source trace; blue is the target coastline. Coverage is not a catch/PPR multiplier. [Geographic assessment](#) | [Native trace controls](#)

